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Week 6

→ it is necessary the case that all α is ϕ ,

Q1. (i) $\models_{\text{S5ML}} \Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ for all α , α is necessarily ϕ

ATC $V_{M,g}(\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi, u) = 0$

$$V_{M,g}(\Box \forall \alpha \phi, u) = 1 \quad \& \quad V_{M,g}(\forall \alpha \Box \phi, u) = 0$$

(1) (2)

(1): $V_{M,g}(\forall \alpha \phi, v) = 1$, for any $v \in W$

for each world $v \in W$,

for any $d \in D$, $V_{M,g_d}(\phi, v) = 1$

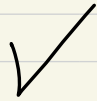
(2): $V_{M,g_u}(\Box \phi, u) = 0$ for some $u \in D$ at world u

$\Rightarrow V_{M,g_u}(\phi, v') = 0$ for some $u \in D$ at some world $v' \in W$

from (1) then, we can choose some world $v = v'$ & $d = u$

s.t $V_{M,g_u}(\phi, v') = 1$

but $V_{M,g_u}(\phi, v') = 0$ ~~X~~



$$(b) \models_{SQML} \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$$

$$\text{AFC that } V_{M,g}(\forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi, u) = 0$$

$$V_{M,g}(\forall \alpha \Box \phi, u) = 1, \quad V_{M,g}(\Box \forall \alpha \phi, u) = 0$$

(1)
(2)

$$(1): \text{for every } d \in D, V_{M,g_d}^\alpha(\Box \phi, u) = 1$$

for each $d \in D$

$$\Rightarrow V_{M,g_d}^\alpha(\phi, v) = 1 \text{ for each } v \in W$$

$$(2) V_{M,g}(\forall \alpha \phi, v') = 0 \text{ for some } v' \in W$$

$$V_{M,g_u}^\alpha(\phi, v') = 0 \text{ for some } u \in D$$

$$\text{From (1), we can choose a } v = v' \text{ and a } d = u$$

$$\text{s.t. } V_{M,g_u}^\alpha(\phi, v') = 1$$

$$\text{but from (2), } V_{M,g_u}^\alpha(\phi, v') = 0 \quad \text{X}$$

✓

(c) $\nexists_{\text{SQL}} \models \exists \alpha \phi \rightarrow \exists \alpha \models \phi$

ω_1

$\begin{array}{c} * \\ \\ \exists \alpha \phi \\ + \end{array}$	$\begin{array}{c} + \\ \\ \exists \alpha \models \phi \end{array}$	$\begin{array}{c} \\ \phi_{g_u}^{\alpha} \\ \\ \exists \phi_{g_v}^{\alpha} \\ * \end{array}$
$\phi: \{u\}$		

$D: \{u, v\}$

ω_2

$\begin{array}{c} \circ \\ \\ \phi_{g_u}^{\alpha} \end{array}$	$\begin{array}{c} \\ \exists \alpha \phi \\ + \end{array}$	$\begin{array}{c} \\ \phi_{g_v}^{\alpha} \end{array}$
$\phi: \{v\}$		

ω_3

$\begin{array}{c} \circ \\ \\ \phi_{g_v}^{\alpha} \end{array}$	$\begin{array}{c} \\ \exists \alpha \phi \end{array}$	$\begin{array}{c} \\ \phi_{g_u}^{\alpha} \end{array}$
$\phi: \{u\}$		

$\mathcal{W} = \{\omega_1, \omega_2, \omega_3\}$

$D = \{u, v\}$, let $\phi = r_2$

$I(F) = \{\langle u, r \rangle, \langle u, b \rangle, \langle v, a \rangle\}$

There doesn't
match
✓

(You're still correct tho)

$$(d) \models_{\text{SQML}} \exists \alpha \Box \phi \rightarrow \Box \exists \alpha \phi$$

$$\text{AFC } V_{M,g}(\exists \alpha \Box \phi \rightarrow \Box \exists \alpha \phi, u) = 0$$

$$V_{M,g}(\exists \alpha \Box \phi, u) = 1 \quad (1), \quad V_{M,g}(\Box \exists \alpha \phi, u) = 0 \quad (2)$$

$$(1): V_{M,g_u}(\Box \phi, u) = 1 \text{ for some } u \in D$$

$$\therefore V_{M,g_u}(\phi, v) = 1 \text{ for all } v \in W$$

$$(2) V_{M,g}(\exists \alpha \phi, v') = 0 \text{ for some } v' \in W$$



$$V_{M,g_d}(\phi, v') = 0 \text{ for all } d \in D$$

$$\text{We can choose a } d = u \text{ st. } V_{M,g_u}(\phi, v') = 0 \text{ in (2)}$$

$$\text{We can choose a } v = v' \text{ st. } V_{M,g_u}(\phi, v') = 1 \text{ in (1)}$$

Contradiction $\times$

$$Q2. (a) i. (P_x)(x/x) = P_x \quad \checkmark$$

$$ii. (P_x \wedge Q_y)(x/y) = (P_x)(x/x) \wedge (Q_y)(x/y)$$

The substitution is only defined on $\neg$ and $\rightarrow$, of which $\wedge$ is an abbreviation.

$$= P_x \wedge \Box((Q_y)(x/y))$$

$$= P_x \wedge \Box Q_x \quad \checkmark$$

$$iii. ((Q_y)(x/y))(z/x) = (Q_x)(z/x) \quad \checkmark$$

$$= Q_z$$

$$iv. (\Box \forall x P_x)(y/x) = \Box(\forall x P_x)(y/x) \quad \checkmark$$

$$= \Box \forall x P_x$$

$$v. (\Box \forall x P_y)(x/y) = \Box(\forall x P_y)(x/y) \quad \checkmark$$

$$= \text{undefined}$$

Since y occurs free in the subformula $\forall x P_y$ that begins with $\forall x$, x is not substitutable for y .

$$\text{vi. } (\forall x R_x \wedge R_{xy})(y/x)$$

$$= (\forall x R_x)(y/x) \wedge (R_{xy})(y/x)$$

$$= \forall x R_x \wedge R_{yy}$$

(b) The definition of $\phi(B/\alpha)$ fits Sider's correct substitution

since (1) for $\phi(B/\alpha)$, only free occurrences of α can be changed ($\therefore \forall \alpha \phi(B/\alpha) = \forall \alpha \phi$), and

(2) all free instances of α must be changed to β

$$(\therefore \Pi^n(\gamma_1, \dots, \gamma_n)(B/\alpha) = \Pi^n(\gamma_1(B/\alpha), \dots, \gamma_n(B/\alpha)))$$

and (3) none of the instantiated occurrences i.e. β appearing from replacing α can be bound in the axiom (so β is not substitutable for α iff α occurs freely in any subformula of ϕ that begins with $\forall \beta$)

Since our definition satisfies Sider's three restrictions,
it is a "correct substitution"

Q3. (a) $\vdash_{\text{SQML}} \forall x f_x \rightarrow \forall y f_y$

1. $\forall x f_x \rightarrow f_y$ PC1
2. $\forall y (\forall x f_x \rightarrow f_y)$ UG
3. $\forall y (\forall x f_x \rightarrow f_y) \rightarrow (\forall x f_x \rightarrow \forall y f_y)$ PC2 ✓
4. $\forall x f_x \rightarrow \forall y f_y$ MP2,3

(b) $\vdash_{\text{SQML}} \forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \forall \alpha \psi)$

1. $\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)$ PC1 ✓
2. $\forall \alpha \phi \rightarrow \phi$ PL
3. $(\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)) \rightarrow ((\forall \alpha \phi \rightarrow \phi) \rightarrow (\forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \psi)))$ PL
4. $\forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \psi)$ MP1,2,3
5. $\forall \alpha (\forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \psi))$ UG4
6. $\forall \alpha (\forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \psi)) \rightarrow (\forall \alpha (\phi \rightarrow \psi) \rightarrow \forall \alpha (\forall \alpha \phi \rightarrow \psi))$ PC2
7. $\forall \alpha (\phi \rightarrow \psi) \rightarrow \forall \alpha (\forall \alpha \phi \rightarrow \psi)$ MP5,6
8. $\forall \alpha (\forall \alpha \phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \forall \alpha \psi)$ PC2 ✓
9. $\forall \alpha (\phi \rightarrow \psi) \rightarrow (\forall \alpha \phi \rightarrow \forall \alpha \psi)$ MP7,8

$$A \wedge B = \sim(\sim A \vee \sim B) = \sim(A \rightarrow \sim B)$$

$$(c) \vdash_{SQML} (\Box \forall \alpha \phi \wedge \Diamond \exists \alpha \psi) \rightarrow \Diamond \exists \alpha (\phi \wedge \psi)$$

$$(1) \vdash_{SQML} \sim(\Box \forall \alpha \phi \rightarrow \sim \Diamond \exists \alpha \psi) \rightarrow \Diamond \exists \alpha \sim(\phi \rightarrow \sim \psi) \quad \text{From (1) to (11) you are doing some double}$$

$$(11) \vdash_{SQML} \sim(\Box \forall \alpha \phi \rightarrow \Box \forall \alpha \sim \psi) \rightarrow \sim \Box \forall \alpha (\phi \rightarrow \sim \psi) \quad \text{negation elimination,}$$

while is not entirely obvious that you can do

$$1. \forall \alpha (\phi \rightarrow \sim \psi) \rightarrow (\forall \alpha \phi \rightarrow \forall \alpha \sim \psi) \quad \text{From (6)}$$

$$2. \Box \forall \alpha (\phi \rightarrow \sim \psi) \rightarrow \Box (\forall \alpha \phi \rightarrow \forall \alpha \sim \psi) \quad \text{NEC, K, MP}$$

$$3. \Box (\forall \alpha \phi \rightarrow \forall \alpha \sim \psi) \rightarrow (\Box \forall \alpha \phi \rightarrow \Box \forall \alpha \sim \psi) \quad \text{K}$$

$$4. \Box \forall \alpha (\phi \rightarrow \sim \psi) \rightarrow (\Box \forall \alpha \phi \rightarrow \Box \forall \alpha \sim \psi) \quad \text{PL (syllogism) 2, 3}$$

$$5. \sim(\Box \forall \alpha \phi \rightarrow \Box \forall \alpha \sim \psi) \rightarrow \sim \Box \forall \alpha (\phi \rightarrow \sim \psi) \quad \text{PL (contradiction)}$$

$$6. \sim(\Box \forall \alpha \phi \rightarrow \sim \Diamond \exists \alpha \psi) \rightarrow \Diamond \exists \alpha \sim(\phi \rightarrow \sim \psi)$$

$$7. (\Box \forall \alpha \phi \wedge \Diamond \exists \alpha \psi) \rightarrow \Diamond \exists \alpha (\phi \wedge \psi)$$

By def of $\exists \alpha, \Diamond$
By def of $\wedge$

Also some
double negation
going on here



$$(d) \vdash_{SQML} \Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$$

1. $\forall \alpha \phi \rightarrow \phi(\alpha/\phi)$ PC1
2. $\Box \forall \alpha \phi \rightarrow \Box \phi$ NEC, K, MP1
3. $\forall \alpha (\Box \forall \alpha \phi \rightarrow \Box \phi)$ UG2
4. $\forall \alpha (\Box \forall \alpha \phi \rightarrow \Box \phi) \rightarrow (\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi)$ PC2
5. $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ MP3,4

✓

$$(e) \vdash_{SQML} \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$$

Lemma: The B axiom is provable in SQML i.e. $\vdash_{SQML} \Diamond \Box \phi \rightarrow \phi$

1. $\Box \phi \rightarrow \phi$ T
2. $\Diamond \Box \phi \rightarrow \Box \phi$ SS
3. $\Diamond \Box \phi \rightarrow \phi$ ~~MP~~ PL

- Pf. 1. $\forall \alpha \Box \phi \rightarrow \Box \phi$ PC1
2. $\Diamond \forall \alpha \Box \phi \rightarrow \Diamond \Box \phi$ I, NEC, K, MP
 3. $\Diamond \Box \phi \rightarrow \phi$ B ~~PL~~
 4. $\Diamond \forall \alpha \Box \phi \rightarrow \phi$ ~~MP~~ 2,3
 5. $\Diamond \forall \alpha \Box \phi \rightarrow \forall \alpha \phi$ 4, UG, PC2, MP
 6. $\Box \Diamond \forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$ 5, B_{der}
 7. $\forall \alpha \Box \phi \rightarrow \Box \Diamond \forall \alpha \Box \phi$ B $\Diamond$
 8. $\forall \alpha \Box \phi \rightarrow \Box \forall \alpha \phi$ ~~MP~~ 6,7
PL

✓

$$(f) \vdash_{\text{SQML}} \forall \alpha \Box \beta (B = \alpha)$$

$$1. \alpha = \alpha \quad \text{RX}$$

$$2. \forall \beta \sim B = \alpha \rightarrow \sim \alpha = \alpha \quad \text{PC1}$$

$$3. \alpha = \alpha \rightarrow \sim \forall \beta \sim B = \alpha \quad \text{PL}$$

$$4. \sim \forall \beta \sim B = \alpha \quad \text{MP1, 3}$$

$$5. \forall \alpha \sim \forall \beta \sim B = \alpha \quad \text{UG}$$

$$6. \forall \alpha \Box \beta (B = \alpha) \quad \text{def of } \Box \beta$$

Q4. Claim: $\vdash_{\text{SQML}} \phi$ iff $\models_{\text{SQML}} \phi$

We have to prove 2 lemmas:

Lemma 1: SQML-axioms are SQML-valid

Lemma 2: SQML-rules preserve SQML-validity

Lemma 1: SQML-axioms are SQML-valid

PL1-PL3 are all PL-valid by previous sheets, and hence S-valid, and SQML valid

K, T, SS axioms are all S-valid by previous sheets, and hence SQML valid

} we can generalise S-validity to SQML since we can treat a MPL-iff as an n-place predicate with only constants as terms

It remains to show that PC1, PC2, RX, II are valid

$$(PC1) \vdash_{\text{SQML}} \forall \alpha \phi \rightarrow \phi(B/\alpha)$$

$$\text{AFC that } V_{\text{Mg}}(\forall \alpha \phi \rightarrow \phi(B/\alpha), u) = 0$$

$$V_g(\forall \alpha \phi, u) = 1, V_g(\phi(B/\alpha), u) = 0$$

$$\text{since } V_{g_{\text{Mg}}}(\phi(B/\alpha), u) = V_{g_{\text{Mg}}}(\phi, u) = 0$$

$$\text{let } [\alpha]_{g_{\text{Mg}}} = [B]_g$$

$$V_g(\forall \alpha \phi, u) = 1 \text{ iff for all } d \in D, V_{g_d}(\phi, u) = 1$$

$$\text{if } [\alpha]_g = [B]_g, \text{ then } V_g(\phi, u) = V_g(\phi(B/\alpha), u)$$

$$\therefore V_{g_{\text{Mg}}}(\phi, u) = 1 \quad \text{X}$$

$$(PC2) \models_{SRML} \forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)$$

$$\text{AFC that } V_{m,g}(\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi), u) = 0$$

$$\underset{(1)}{V_g(\forall \alpha (\phi \rightarrow \psi), u)} = 1, \underset{(2)}{V_g(\phi \rightarrow \forall \alpha \psi, u)} = 0$$

$$(1) V_g(\forall \alpha (\phi \rightarrow \psi), u) = 1 \text{ iff for all } d \in D$$

$$V_{g_d}(\phi \rightarrow \psi, u) = 1$$

$$(2) V_g(\phi, u) = 1 \text{ and } V_g(\forall \alpha \psi, u) = 0$$

$$\text{for some } u \in D, V_{g_u}(\psi, u) = 0$$

since α does not occur freely in ϕ ,

$$g \& g_u \text{ agree on free variables, } V_g(\phi, u) = V_{g_u}(\phi, u) = 1$$

$$\therefore V_{g_u}(\phi \rightarrow \psi, u) = 0$$

$$\text{let } d = u, V_{g_u}(\phi \rightarrow \psi, u) = 1 \quad \times$$

$$(RK) \models_{SRML} \alpha = \alpha$$

$$\text{AFC that } V_{m,g}(\alpha = \alpha, u) = 0$$

But any g ,

$$[\alpha]_{m,g} = [\alpha]_{m,g}$$

$$\therefore V_{m,g}(\alpha = \alpha, u) = 1 \quad \times$$

$$(II) \models_{SRML} \alpha = \beta \rightarrow (\phi(\alpha) \rightarrow \phi(\beta))$$

$$\text{AFC that } V_{m,g}(\alpha = \beta \rightarrow (\phi(\alpha) \rightarrow \phi(\beta)), u) = 0$$

$$\therefore \underset{(1)}{V_{m,g}(\alpha = \beta, u)} = 1, \underset{(2)}{V_{m,g}(\phi(\alpha) \rightarrow \phi(\beta), u)} = 0$$

$$(1) : [\alpha]_{m,g} = [\beta]_{m,g} \text{ at some world } w$$

$$(2) V_{m,g}(\phi(\alpha), u) = 1, V_{m,g}(\phi(\beta), u) = 0$$

But for any wff ϕ , this leads to a contradiction
 Consider any non-atomic subformula of ϕ . Call this ψ

if $[\alpha]_{M,g} = [\beta]_{M,g}$, then for any tuple

$$\langle [\alpha_1]_{M,g}, [\alpha_2]_{M,g}, \dots \rangle = \langle [\beta_1]_{M,g}, [\beta_2]_{M,g}, \dots \rangle$$

then, for any non-atomic subformula ψ in ϕ ,

$$V_{M,g}(\psi(\alpha), u) = V_{M,g}(\psi(\beta), u)$$

But if $V_{M,g}(\phi(\alpha), u) \neq V_{M,g}(\phi(\beta), u)$

there is at least one subformula χ in ϕ

$$\text{s.t. } V_{M,g}(\chi(\alpha), u) \neq V_{M,g}(\chi(\beta), u) \quad \text{X}$$

Lemma 2: All SQML rules preserve SQML validity

Since MP preserves PL-validity, it preserves SQML validity

Since NEC preserves SS-validity, it preserves SQML validity

} in Week 4

Remains to show that UG preserves SQML validity

UG: if $\models_{\text{SQML}} \phi$, then $\models_{\text{SQML}} \forall \alpha \phi$

Assume $\models_{\text{SQML}} \phi$

$\therefore V_{M,g}(\phi, u) = 1$ holds for any $u \in U$, for any g in each world, for any model M

Consider any model M ,

we know that for any world u ,

$$V_{M,g}(\phi, u) = 1 \text{ for any } g$$

$$\therefore V_{M,g}^a(\phi, u) = 1 \text{ for any } u \in D$$

$$\therefore V_{M,g}(\forall a \phi, u) = 1$$

Since this holds for any u ,

$$V_{M,g}(\forall a \phi, u) = 1 \text{ for all } u \in W$$

$\therefore$ it is true at some model M

since this holds for all M ,

$$\models_{SQML} \forall a \phi \quad \square$$

Main pf. Let $\vdash_n \phi$ denote a proof of length n . We prove by induction on the length of proof

BC: $\vdash_1 \phi$

By def, ϕ is an SQML axiom

By Lemma 1,

$\vdash \phi$ since SQML axioms are SQML valid

IH: Suppose $\vdash_n \phi$ implies $\vdash \phi$ for all $n < m$, we wish that $\vdash_m \phi$ implies $\vdash \phi$

Suppose $\vdash_m \phi$. There are 4 cases for the last line of ϕ
to obtain

Case 1: ϕ is an SQML axiom

by lemma 1, $\vdash \phi$

Case 2: ϕ is obtained by NEC

Then $\phi = \Box \psi$, $\vdash_n \psi$ with $n < m$

By IH, $\vdash \psi$

since if $\vdash \psi$, then $\vdash \Box \psi$ by Lemma 2,

$\vdash \phi$

Case 3: ϕ is obtained by $\cup\mathcal{L}$

Then $\phi = \cup \alpha \psi$, $\vdash_n \psi$, $n < m$

by IH, $\vdash \psi$

By lemma 2, $\vdash \cup \alpha \psi$

$\therefore \vdash \phi$

Case 4: ϕ is obtained by MP

$\therefore \vdash_{n_1} \psi$ & $\vdash_{n_2} \psi \rightarrow \phi$ with $n_1, n_2 < m$

$\therefore \vdash \psi$ & $\vdash \psi \rightarrow \phi$ by IH

$\therefore \vdash \phi$ by lemma 2

Claim proven i.e. if $\vdash_{\text{SOM}} \phi$, then $\vdash_{\text{SOM}} \phi$

